BEng(Hons) Software Engineering

Module Name	4COSC006C- Software Development -I
Module Leader	Mr. Guhanathan Poravi
Assessment Type	Individual
Assessment Number	Stage 03
Student ID	20231970 – IIT W2149655 – UoW
Student Name	Mr. K. M. Palansuriya
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Test Cases for JSON file operations

Test Cases and Results

Test Case 01 – Loading tasks from JSON

Table 1 - Loading tasks from JSON

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
loading tasks from JSON	Start the program	The program loads tasks from the Json file of "tasks.json" file if it exists. If no file exists. It initializes an empty task list	Tasks are loaded from the tasks.json file. If the file does not exist an empty list is initialized in the json file	Pass

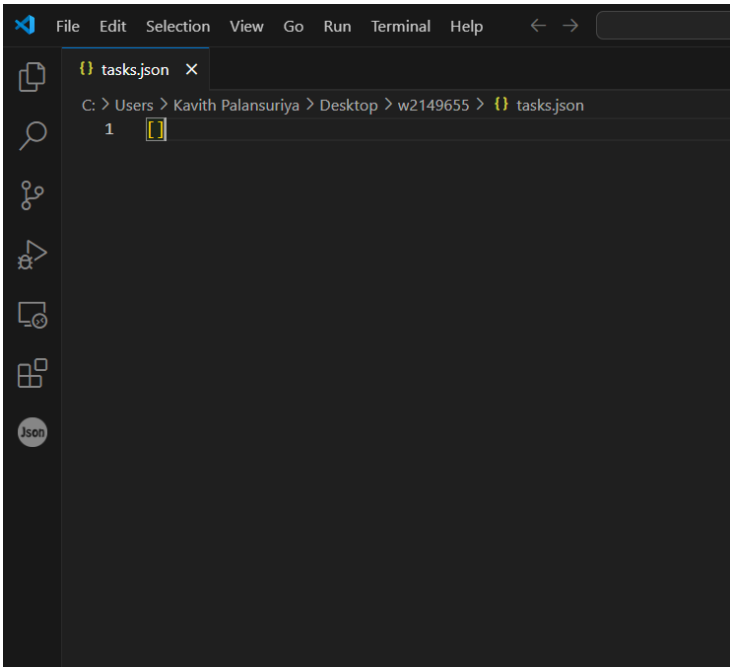


Figure 1 - Loading tasks from JSON (JSON FILE)

Test Case 02 – Adding a new task

Table 2 - Adding a new task

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Adding a new task	Task Name: "Task 01", Description:"SD01 Stage 01,02", Priority: "medium" , Due Date:"2025-03-27"	The task is added to the task list and saved to the JSON file.	Task is added to the list and saved to the file	Pass

```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======= RESTART: C:\Users\Kavith Palansuriya\Desktop\w2149655\Stage_03.py ======
Welcome to Task Manager.

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]: 1
Enter your task name: Task 01
Enter the task description:SD01 Stage 01,02
Enter the priority of the task (high,medium,low): medium
Enter the task due date in the following format (yyyy-mm-dd): 2025-03-27
Task added successfully!

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]:
```

Test Case 1 - Adding a new task (IDLE FILE)

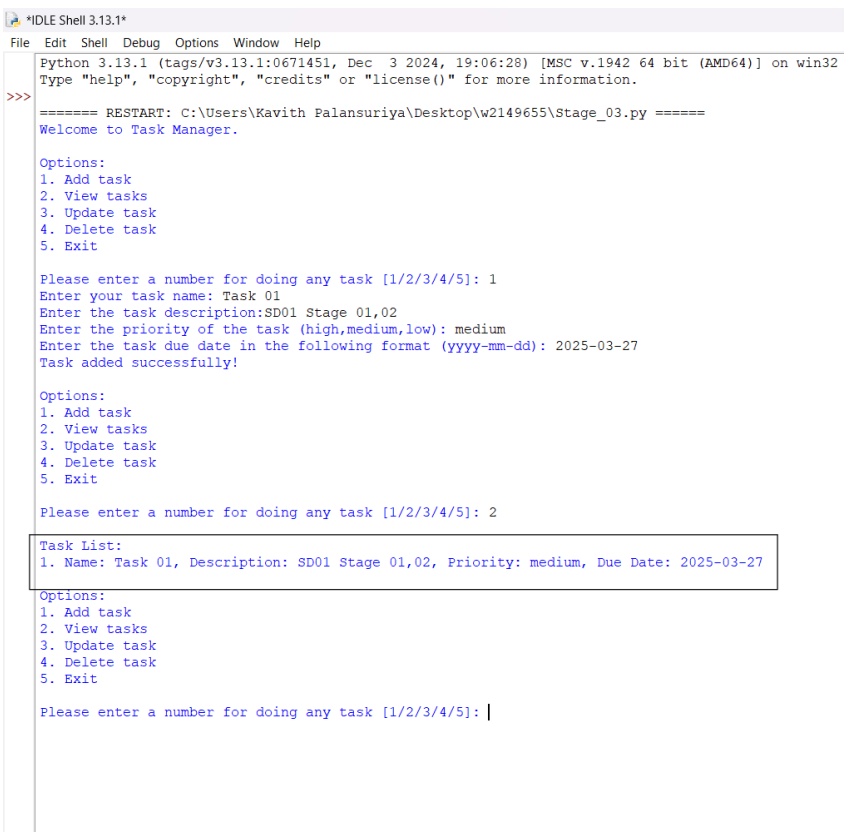
```
File  Edit  Selection  View  Go  Run  Terminal  Help
{} tasks.json
C: > Users > Kavith Palansuriya > Desktop > w2149655 > {} tasks.json > ...
1  [
2      [
3          "Task 01",
4          "SD01 Stage 01,02",
5          "medium",
6          "2025-03-27"
7      ]
8  ]
```

Figure 2- Adding a new task (JSON FILE)

Test Case 03 – View tasks

Table 3 - View tasks

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
View Tasks	After adding a task run the program	The task list is displayed correctly with task details	Tasks are displayed as Expected in the format (Name, description, priority, due date)	Pass



Test Case 2 - View tasks (IDLE FILE)

Test Case 04 – Updating a task

Table 4 - Updating a task

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Updating a task	Task Name: “Task 01”, Description:”SD01 Stage 01,02”, Priority: “medium” , Due Date:”2025-03-27” **Updating this old task into new task** Task Name: “Task 01”, Description:”SD01 Stage 03,04”, Priority: “high” , Due Date:”2025-03-27”	The task is updated with the new name, description, and priority. The changes are reflected in the task list and saved to the JSON file	Task is updated with the new Details, and changes are saved to “tasks.json”	Pass

```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======= RESTART: C:\Users\Kavith Palansuriya\Desktop\w2149655\Stage_03.py ======
Welcome to Task Manager.

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]: 3

Task List:
1. Name: Task 01, Description: SD01 Stage 01,02, Priority: medium, Due Date: 2025-03-27
Enter the task number to update:1
Enter new task name: Task 01
Enter new description: SD01 Stage 03,04
Enter new priority (high/medium/low): high
Task updated successfully!

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]:
```

Test Case 3 - Updating a task (IDLE FILE)

```
File  Edit  Selection  View  Go  Run  Terminal  Help
{} tasks.json ×
C: > Users > Kavith Palansuriya > Desktop > w2149655 > {} tasks.json > ...
1  [
2      [
3          "Task 01",
4          "SD01 Stage 03,04",
5          "high",
6          "2025-03-27"
7      ]
8  ]
```

Figure 3 - Updating a task (JSON FILE)

Test Case 05 – Deleting a Task

Table 5 - Deleting a Task

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Deleting a Task	User to enter option 4 delete task and it shows the list and give Task number 1 to delete	The task is deleted from the task list and the changes are saved to the JSON file	Task is deleted from the list and the changes are saved	Pass

Test Case 4 - Deleting a Task (IDLE FILE)

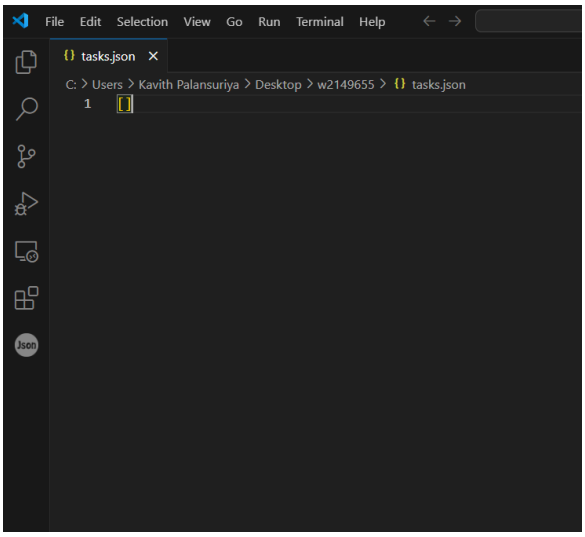
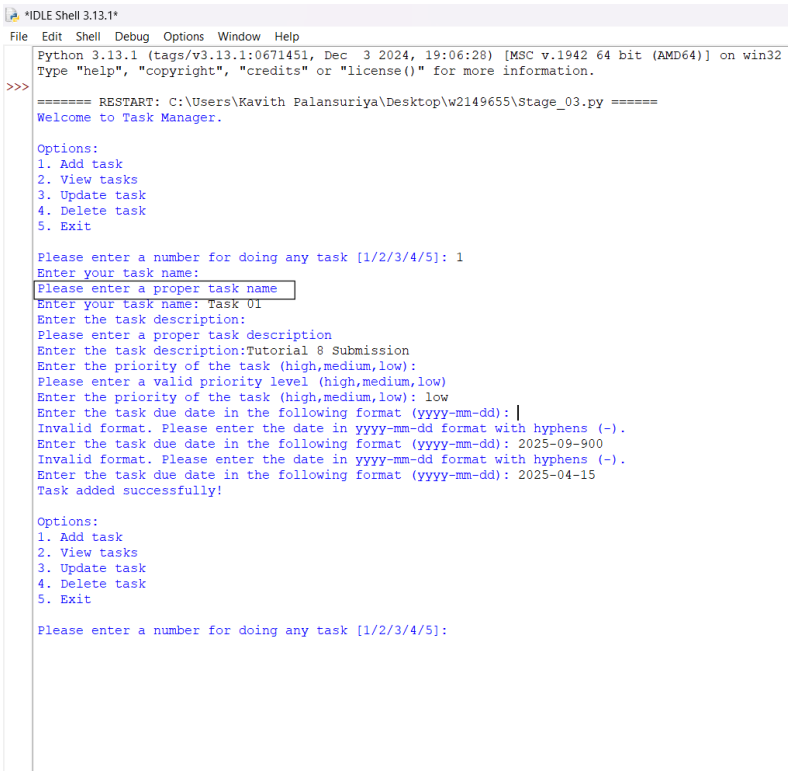


Figure 4 - Deleting a Task (JSON FILE)

Test Case 06 – Adding a task with invalid input of the task name

Table 6 - Adding a task with invalid input of the task name

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Adding a task with invalid input of the task name	Task Name: “”	The system requires the user to enter proper task name	User is prompted to enter a proper task name	Pass



```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======= RESTART: C:\Users\Kavith Palansuriya\Desktop\w2149655\Stage_03.py ======
Welcome to Task Manager.

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]: 1
Enter your task name:
Please enter a proper task name
Enter your task name: Task 01
Enter the task description:
Please enter a proper task description
Enter the task description:Tutorial 8 Submission
Enter the priority of the task (high,medium,low):
Please enter a valid priority level (high,medium,low)
Enter the priority of the task (high,medium,low): low
Enter the task due date in the following format (yyyy-mm-dd):
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-09-900
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-04-15
Task added successfully!

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]:
```

Test Case 5 - Adding a task with invalid input of the task name (IDLE FILE)

Test Case 07 - Adding a task with invalid priority

Table 7 - Adding a task with invalid priority

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Adding a task with invalid priority	Task Name: “Task 01”, Description: “Tutorial 8 Submission”, Priority: “”	The system requires from the user to enter a valid priority (“high”,”medium”,”low”)	User is prompted to enter a valid priority	Pass

```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======= RESTART: C:\Users\Kavith Palansuriya\Desktop\w2149655\Stage_03.py ======
Welcome to Task Manager.

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]: 1
Enter your task name:
Please enter a proper task name
Enter your task name: Task 01
Enter the task description:
Please enter a proper task description
Enter the task description:Tutorial 8 Submission
Enter the priority of the task (high,medium,low):
Please enter a valid priority level (high,medium,low)
Enter the priority of the task (high,medium,low): low
Enter the task due date in the following format (yyyy-mm-dd):
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-09-900
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-04-15
Task added successfully!

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]:
```

Test Case 6 - Adding a task with invalid priority (IDLE FILE)

Test Case 08 - Adding a task with invalid date format

Table 8 - Adding a task with invalid date format

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Adding a task with invalid date format	Task Name: "Task 01", Description: "Tutorial 8 Submission", Priority: "low" , Due Date:"2025-09-900"	The system asks the user to enter a valid due date in this format (yyyy-mm-dd)	User is prompted to enter a valid due date	Pass

```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======= RESTART: C:\Users\Kavith Palansuriya\Desktop\w2149655\Stage_03.py ======
Welcome to Task Manager.

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]: 1
Enter your task name:
Please enter a proper task name
Enter your task name: Task 01
Enter the task description:
Please enter a proper task description
Enter the task description:Tutorial 8 Submission
Enter the priority of the task (high,medium,low):
Please enter a valid priority level (high,medium,low)
Enter the priority of the task (high,medium,low): low
Enter the task due date in the following format (yyyy-mm-dd):
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-09-900
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-04-15
Task added successfully!

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]:
```

Test Case 7 -Adding a task with invalid date format (IDLE FILE)

Test Case 09 - Adding a task with valid input

Table 9 - Adding a task with valid input

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Adding a task with valid input	Task Name: “Task 01”, Description: “Tutorial 8 Submission”, Priority: “low” , Due Date:”2025-04-15”	Task is added successfully and saved to the JSON file	Task is successfully added and saved	Pass

```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======= RESTART: C:\Users\Kavith Palansuriya\Desktop\w2149655\Stage_03.py ======
Welcome to Task Manager.

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]: 1
Enter your task name:
Please enter a proper task name
Enter your task name: Task 01
Enter the task description:
Please enter a proper task description
Enter the task description:Tutorial 8 Submission
Enter the priority of the task (high,medium,low):
Please enter a valid priority level (high,medium,low)
Enter the priority of the task (high,medium,low): low
Enter the task due date in the following format (yyyy-mm-dd):
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-09-900
Invalid format. Please enter the date in yyyy-mm-dd format with hyphens (-).
Enter the task due date in the following format (yyyy-mm-dd): 2025-04-15
Task added successfully!

Options:
1. Add task
2. View tasks
3. Update task
4. Delete task
5. Exit

Please enter a number for doing any task [1/2/3/4/5]:
```

Test Case 8 - Adding a task with valid input (IDLE FILE)

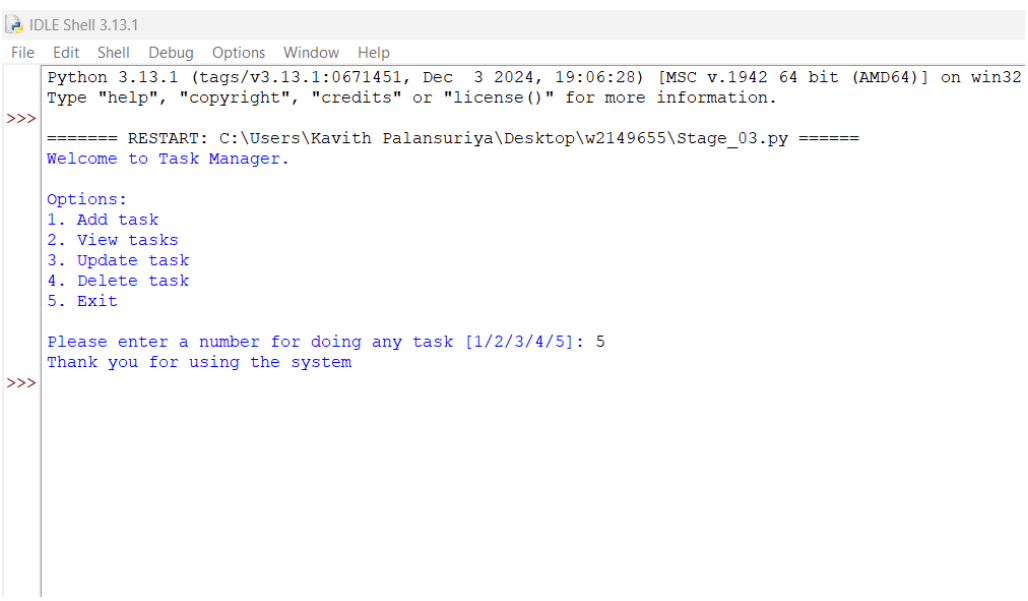
```
File  Edit  Selection  View  Go  Run  Terminal  Help
{} tasks.json ×
C: > Users > Kavith Palansuriya > Desktop > w2149655 > {} tasks.json > ...
1  [
2      [
3          "Task 01",
4          "Tutorial 8 Submission",
5          "low",
6          "2025-04-15"
7      ]
8  ]
```

Figure 5 - Adding a task with valid input (JSON FILE)

Test Case 10 – Exit

Table 10 - Exit

Test Case	Input	Expected Outcomes	Actual Outcomes	Result
Exit	User to enter "option 5"	Program should exit with a massage	Program exited with 'Thank you for using the system'	Pass



Test Case 9 - Exit